

[1] Abstract

This project presents a full-stack intelligent system for **predicting stock price movement** (UP/DOWN) and **evaluating IPO investment potential** using Logistic Regression on 2-year historical OHLCV market data.

A **rule-based IPO scoring engine** rates IPOs using: subscription rate, GMP%, profitability, and sector risk.

The unified web app gives retail investors **AI-assisted, data-driven insights** in real time.

[2] Introduction & Motivation

- Retail investors lack **institutional-grade** tools.
- Volatility causes **sentiment-driven losses**.
- IPO evaluation needs **multiple signals**.
- Existing platforms are complex or expensive.

Objective: Free, accessible ML platform for smarter investment decisions in real time.

- Live **Yahoo Finance** via **yfinance**.
- 10+ NSE, BSE, and NASDAQ tickers.

[3] Dataset & Features

Source: **yfinance** — 2-year daily OHLCV.

Feature	Description
MA5	5-day rolling mean of Close
MA10	10-day rolling mean of Close
Price_Change	$\Delta C_t = C_t - C_{t-1}$
Volume_Change	$\Delta V_t = V_t - V_{t-1}$
Target y	1=UP, 0=DOWN (next day)

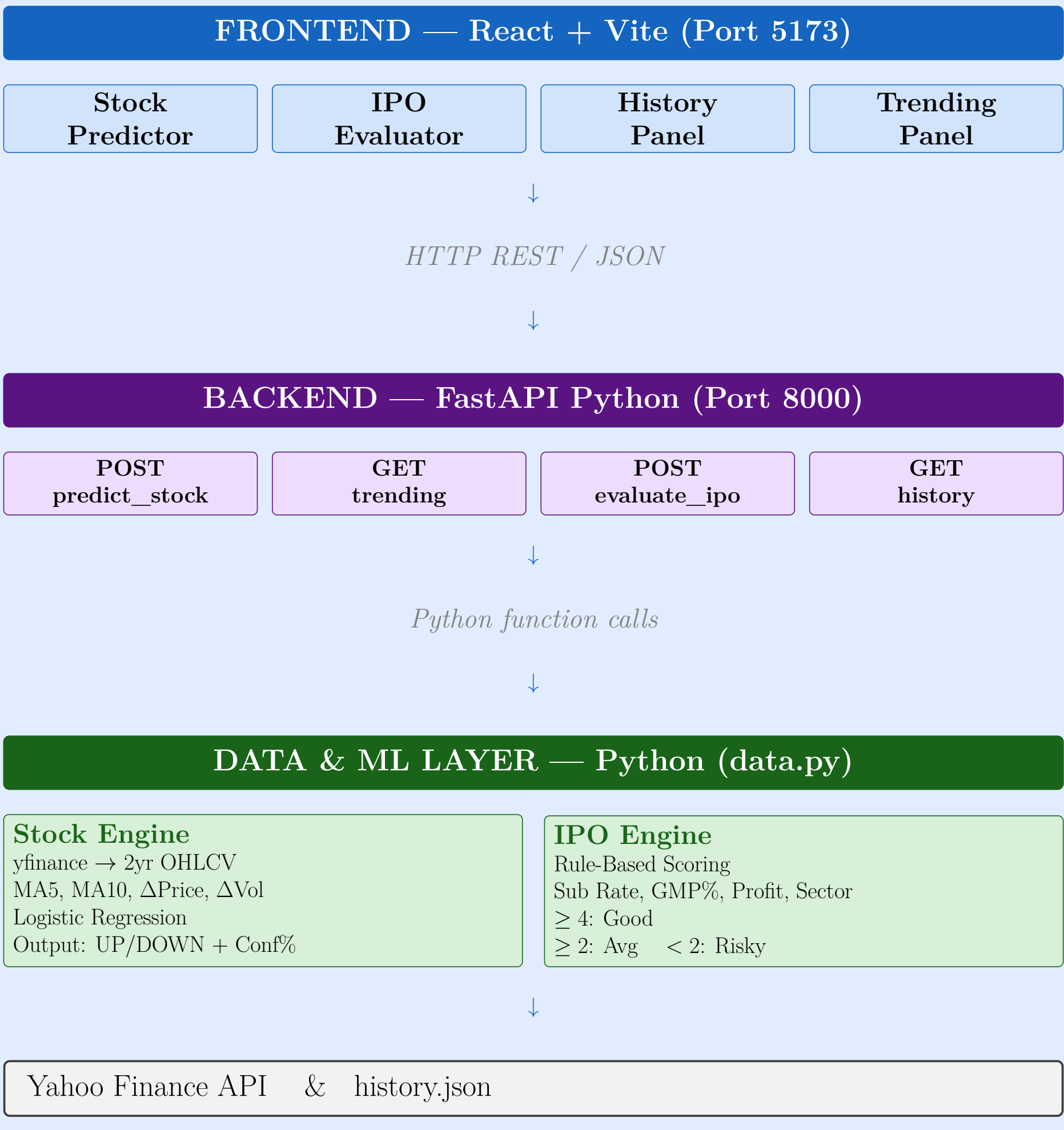
Preprocessing:

- Drop NaN rows after rolling windows
- StandardScaler** normalization
- 80/20** chronological split

[4] Technology Stack

Layer	Technology
Frontend	React 18, Vite 5, CSS
Backend	FastAPI, Python 3.11
ML Model	scikit-learn (LogReg)
Data	pandas, NumPy
Market	yfinance API
Format	JSON / HTTP REST
Storage	history.json

[5] System Architecture



[6] Methodology

A. Stock Prediction:

- Fetch 2yr OHLCV via **yfinance**
- Compute MA5, MA10, ΔPrice, ΔVolume
- Target: $close_{t+1} > close_t \Rightarrow y = 1$
- Normalize with **StandardScaler**
- Train **Logistic Regression** on 80% data
- Return: Prediction, Confidence%, Risk Level

B. IPO Scoring:

Signal	Condition	Pts
Sub Rate	$> 20 \times / 10-20 \times / < 10 \times$	+2/ +1/ -1
GMP%	$> 25\% / 10-25\% / < 10\%$	+2/ +1/ -1
Profit	Profitable / Loss	+1/ -1
Sector	Strong/Neutral/Risky	+1/0/ -1
Rating		≥ 4:Good ≥ 2:Avg < 2:Risky

[7] Results & Performance

Accuracy ~ 55% Logistic Reg.	Confidence ~ 68% Predict Proba.
Stocks 10+ NSE, BSE, NASDAQ	IPO Signals 4 Sub, GMP, Profit, Sec

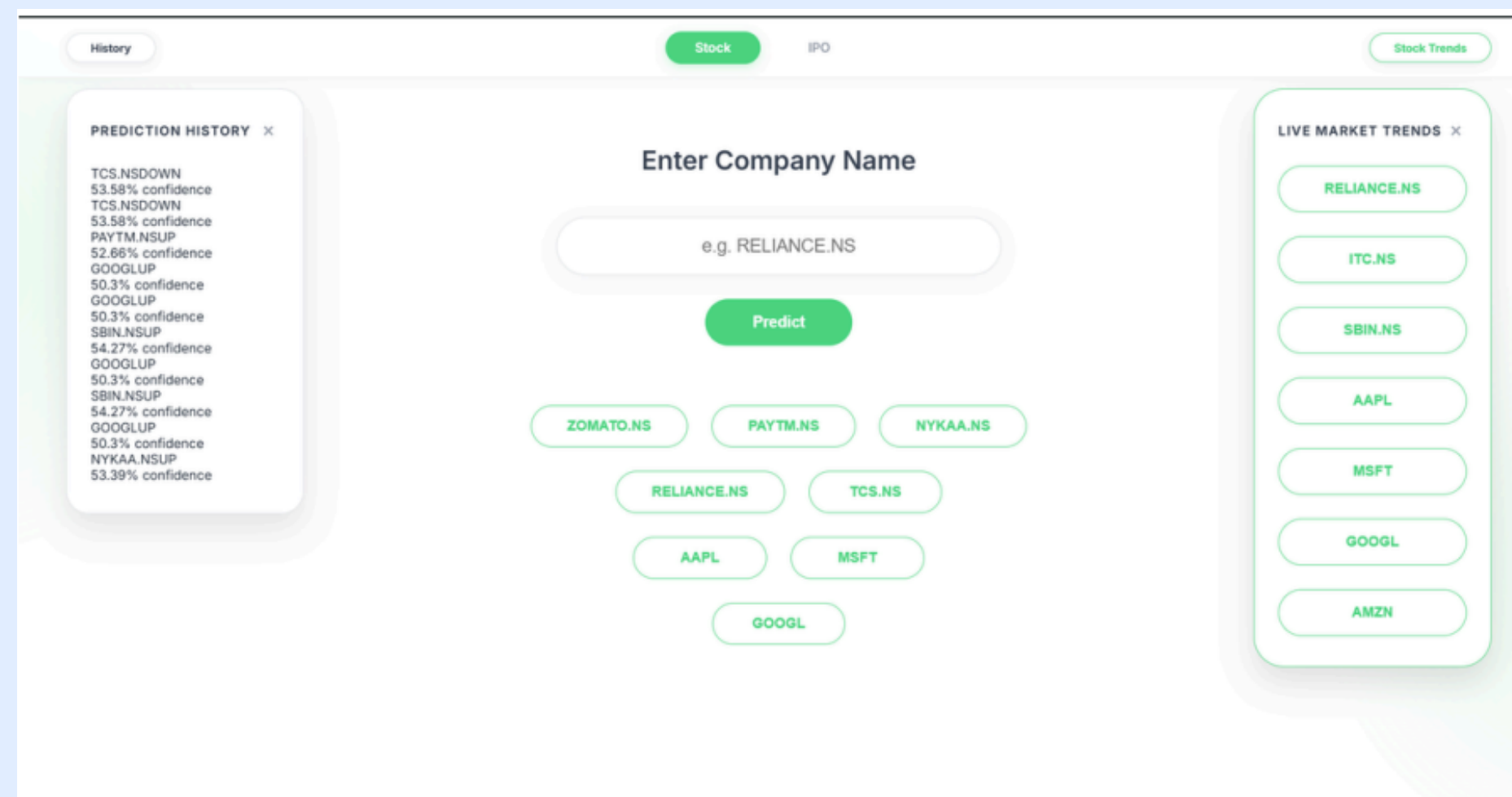
Metric	Score	Note
Accuracy	55%	+5% over baseline
Precision	68%	UP reliability
Recall	74%	UP movements
F1	71%	Balanced

Learned Model Weights (RELIANCE.NS sample):

Feature	Weight (β)	Effect
Intercept	+0.0624	Base UP bias
MA5	-0.3868	High MA5 → DOWN
MA10	+0.2565	High MA10 → UP
Price_Change	-0.0073	Weak momentum
Volume_Change	+0.1292	Rising vol → UP

Key: MA5 is always -ve (mean-reversion), MA10 always +ve (trend-following). Weights differ per ticker.

[8] Application UI — Live Demo



Stock Tab: User enters a ticker (e.g. RELIANCE.NS), clicks Predict — receives UP/DOWN verdict, confidence %, risk level, and 4 plain-language reasons. **Panels:** History (left) and Live Market Trends (right) update automatically.

[9] Conclusion & Future Work

Conclusions:

- Full-stack ML app built and deployed.
- ~55% accuracy on 4 technical features.
- Interpretable, reason-backed IPO ratings.
- Real-time Yahoo Finance predictions.

Future Work:

- LSTM / Transformer for temporal patterns.
- News & social sentiment analysis.
- ML-based IPO model on historical data.
- Cloud (AWS/GCP) + user authentication.

[10] References

- Pedregosa et al. (2011). scikit-learn. *JMLR* 12.
- yfinance**. pypi.org/project/yfinance
- FastAPI. fastapi.tiangolo.com
- React. react.dev
- Breiman (2001). *Mach. Learn.* 45.